

PRODIGY INFOTECH

DATA SCIENCE INTERNSHIP - TASK 05

US Traffic Accident Data Analysis

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Abstract

This project analyzes the US Accidents dataset to identify patterns related to road conditions, weather, time of day, and accident severity using exploratory data analysis and visualization.

Dataset Preview

	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	End_Lat	End_Lng	Distance(mi)	...	Roundabout	Stat
0	A-1	Source2	3	2016-02-08 05:46:00	2016-02-08 11:00:00	39.865147	-84.058723	NaN	NaN	0.01	...	False	F
1	A-2	Source2	2	2016-02-08 06:07:59	2016-02-08 06:37:59	39.928059	-82.831184	NaN	NaN	0.01	...	False	F
2	A-3	Source2	2	2016-02-08 06:49:27	2016-02-08 07:19:27	39.063148	-84.032608	NaN	NaN	0.01	...	False	F
3	A-4	Source2	3	2016-02-08 07:23:34	2016-02-08 07:53:34	39.747753	-84.205582	NaN	NaN	0.01	...	False	F
4	A-5	Source2	2	2016-02-08 07:39:07	2016-02-08 08:09:07	39.627781	-84.188354	NaN	NaN	0.01	...	False	F

Dataset Information

on	Stop	Traffic_Calming	Traffic_Signal	Turning_Loop	Sunrise_Sunset	Civil_Twilight	Nautical_Twilight	Astronomical_Twilight
se	False		False	False	Night	Night	Night	Night
se	False		False	False	Night	Night	Night	Day
se	False		False	True	False	Night	Night	Day
se	False		False	False	False	Night	Day	Day
se	False		False	True	False	Day	Day	Day

Missing Values

```
[6]: df.shape
[6]: (50000, 46)

[7]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 46 columns):
#   Column              Non-Null Count  Dtype
---  -
0    ID                   50000 non-null  object
1    Source               50000 non-null  object
2    Severity             50000 non-null  int64
3    Start_Time           50000 non-null  object
4    End_Time             50000 non-null  object
5    Start_Lat            50000 non-null  float64
6    Start_Lng            50000 non-null  float64
7    End_Lat              0 non-null      float64
8    End_Lng              0 non-null      float64
9    Distance(mi)         50000 non-null  float64
10   Description          50000 non-null  object
11   Street               50000 non-null  object
12   City                 50000 non-null  object
13   County               50000 non-null  object
14   State                50000 non-null  object
15   Zipcode              49993 non-null  object
16   Country              50000 non-null  object
17   Timezone             49993 non-null  object
18   Airport_Code         49993 non-null  object
19   Weather_Timestamp    49734 non-null  object
20   Temperature(F)       49544 non-null  float64
21   Wind_Chill(F)        2497 non-null   float64
22   Humidity(%)          49361 non-null  float64
23   Pressure(in)         49694 non-null  float64
24   Visibility(mi)       49474 non-null  float64
25   Wind_Direction       49733 non-null  object
26   Wind_Speed(mph)      41578 non-null  float64
27   Precipitation(in)    3276 non-null   float64
28   Weather_Condition    49397 non-null  object
29   Amenity              50000 non-null  bool
```

Severity Distribution

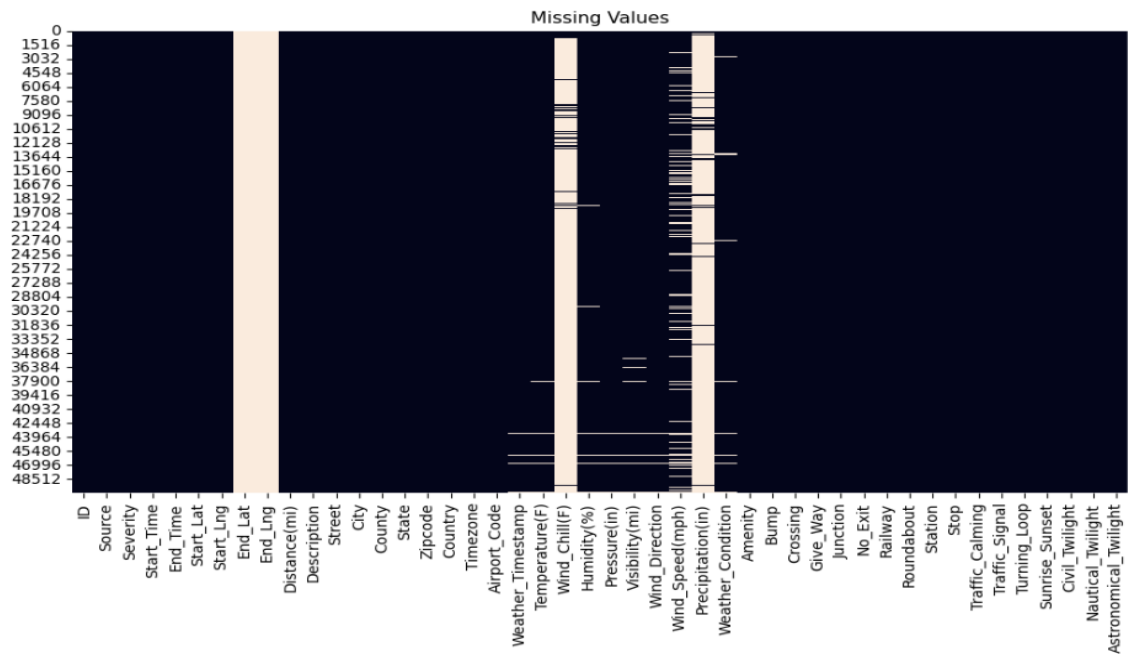
```
[8]: df.describe()

[8]:
```

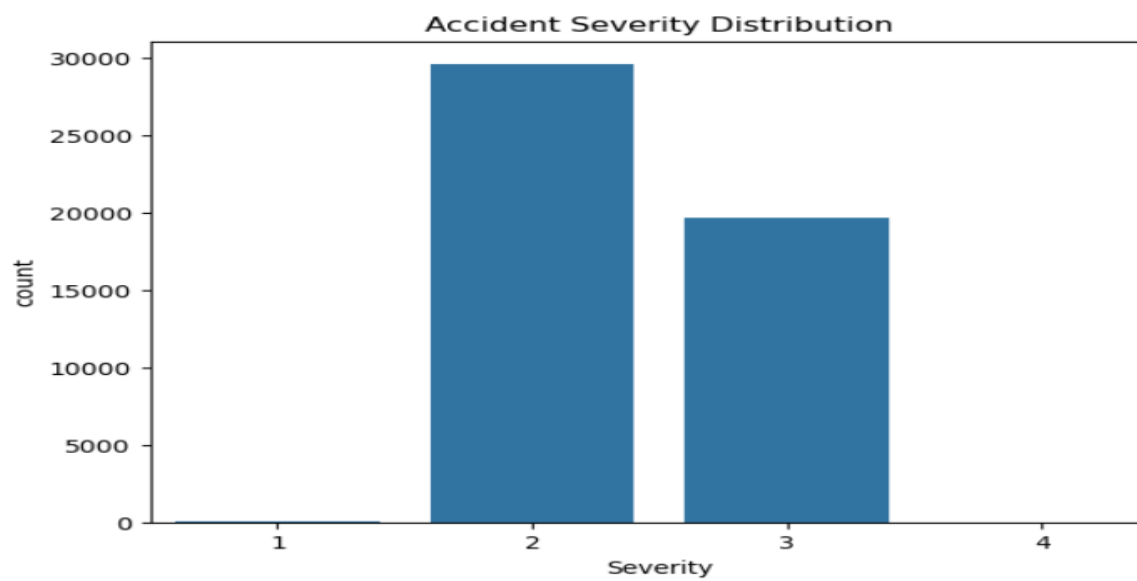
	Severity	Start_Lat	Start_Lng	End_Lat	End_Lng	Distance(mi)	Temperature(F)	Wind_Chill(F)	Humidity
count	50000.00000	50000.000000	50000.000000	0.0	0.0	50000.000000	49544.000000	2497.000000	49361.000000
mean	2.39678	37.396862	-120.736826	NaN	NaN	0.014660	64.975323	39.050541	61.715000
std	0.49225	1.553001	4.750326	NaN	NaN	0.354411	13.200795	14.914320	21.706000
min	1.00000	32.544304	-123.813927	NaN	NaN	0.000000	3.900000	-8.400000	4.000000
25%	2.00000	37.343647	-122.127098	NaN	NaN	0.000000	55.900000	33.400000	46.000000
50%	2.00000	37.776711	-121.828537	NaN	NaN	0.000000	64.400000	39.000000	63.000000
75%	3.00000	38.409191	-121.272102	NaN	NaN	0.010000	73.400000	42.500000	78.000000
max	4.00000	41.428753	-81.550728	NaN	NaN	51.130000	113.000000	101.000000	100.000000

```
[9]: df.columns
Index(['ID', 'Source', 'Severity', 'Start_Time', 'End_Time', 'Start_Lat',
       'Start_Lng', 'End_Lat', 'End_Lng', 'Distance(mi)', 'Description',
       'Street', 'City', 'County', 'State', 'Zipcode', 'Country', 'Timezone',
       'Airport_Code', 'Weather_Timestamp', 'Temperature(F)', 'Wind_Chill(F)',
       'Humidity(%)', 'Pressure(in)', 'Visibility(mi)', 'Wind_Direction',
       'Wind_Speed(mph)', 'Precipitation(in)', 'Weather_Condition', 'Amenity',
       'Bump', 'Crossing', 'Give_Way', 'Junction', 'No_Exit', 'Railway',
       'Roundabout', 'Station', 'Stop', 'Traffic_Calming', 'Traffic_Signal',
       'Turning_Loop', 'Sunrise_Sunset', 'Civil_Twilight', 'Nautical_Twilight',
       'Astronomical_Twilight'],
      dtype='object')
```

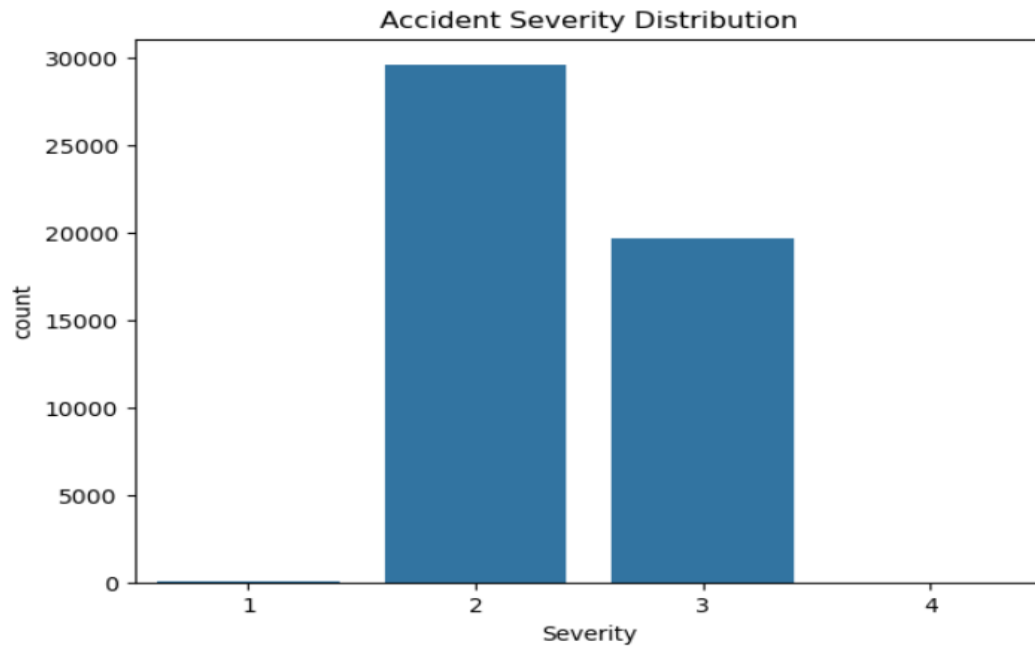
Accidents by Hour



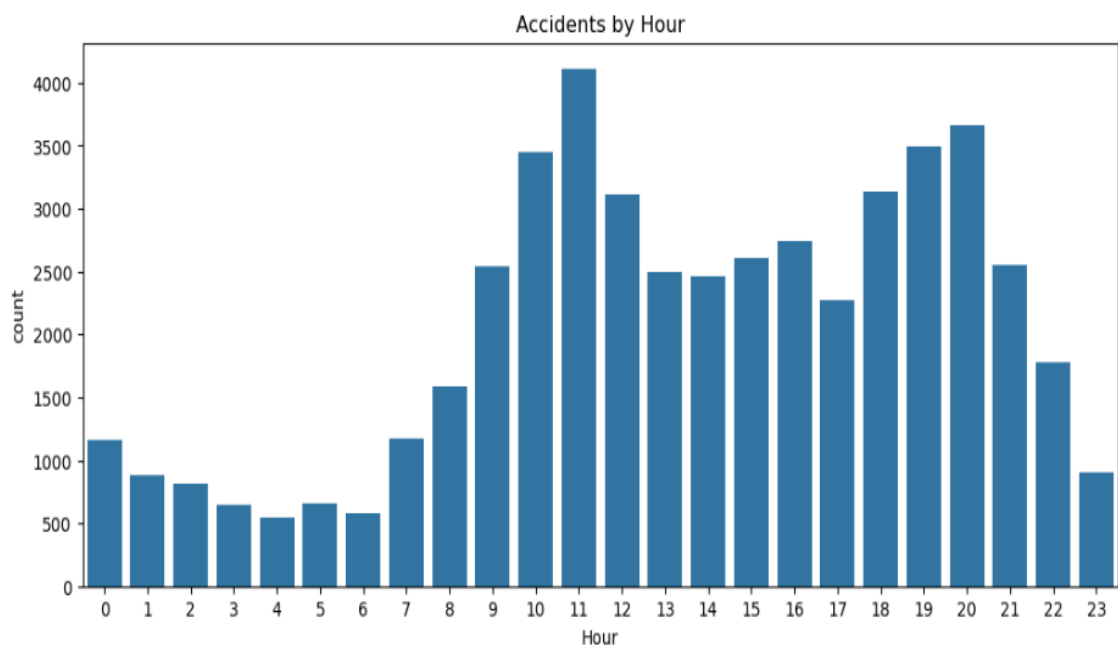
Accidents by Day



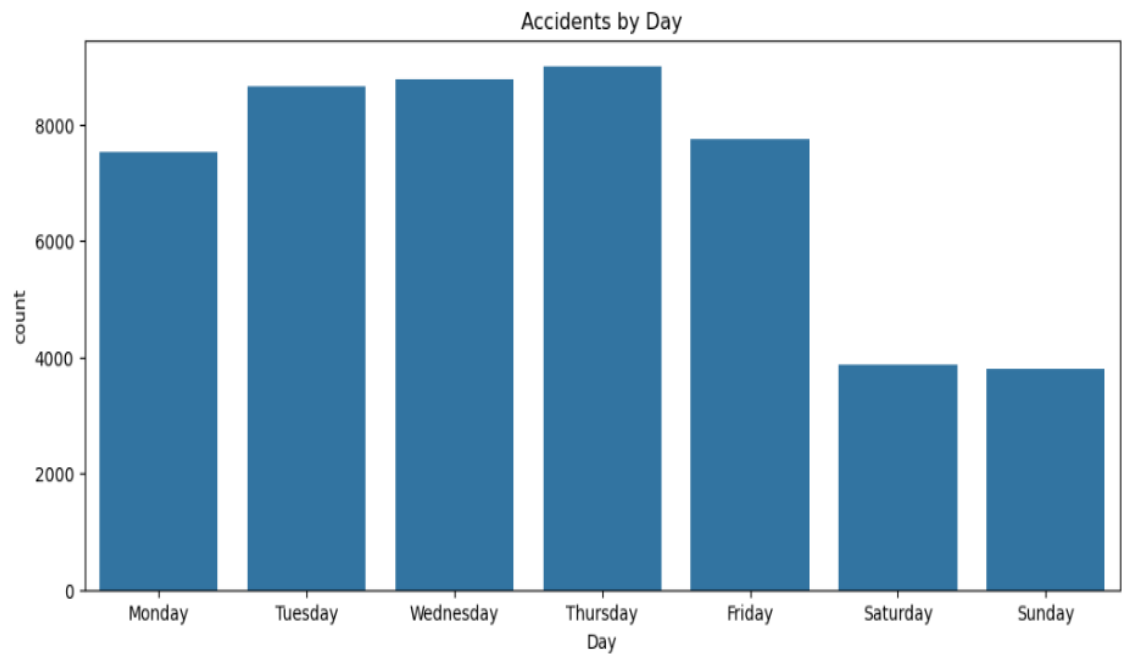
Top States



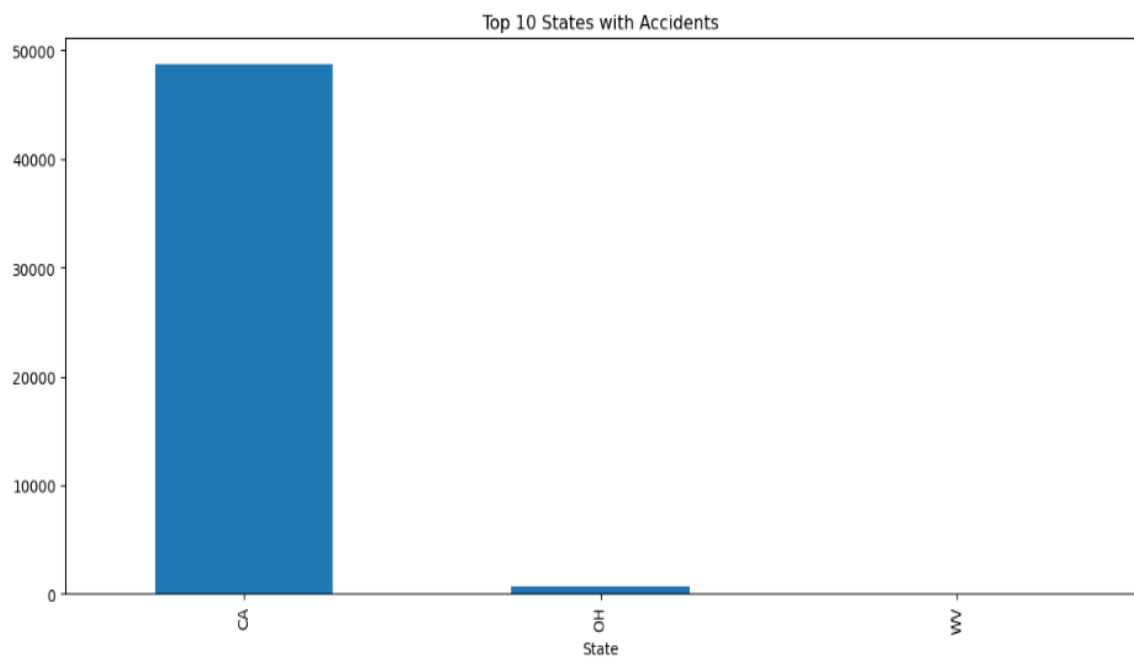
Weather Conditions



Correlation Heatmap



Temperature vs Severity



Visibility vs Severity

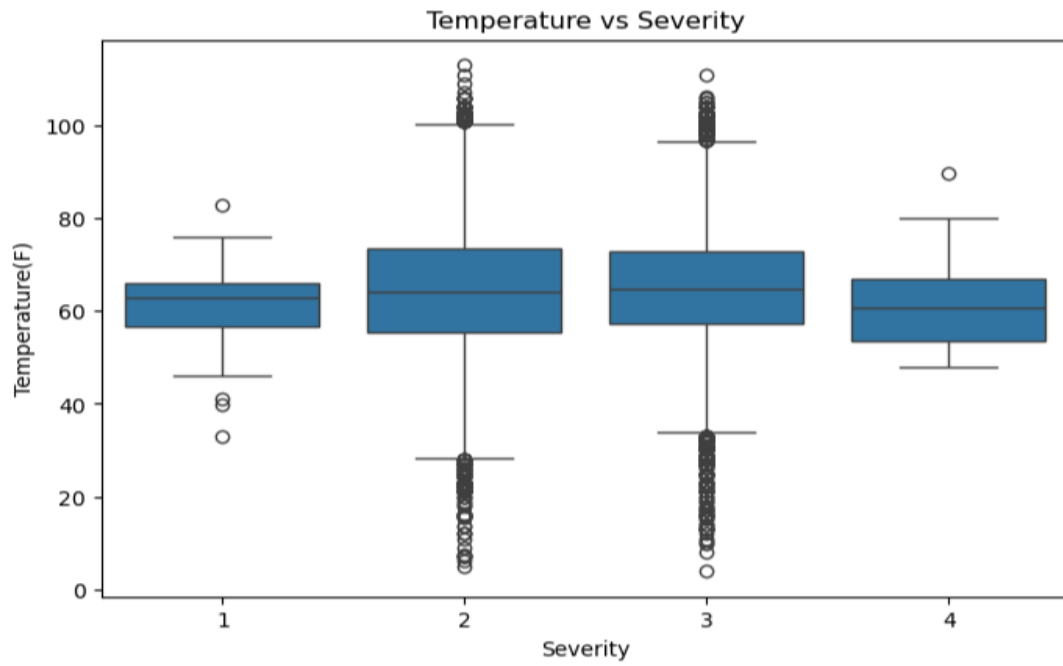


Figure 14

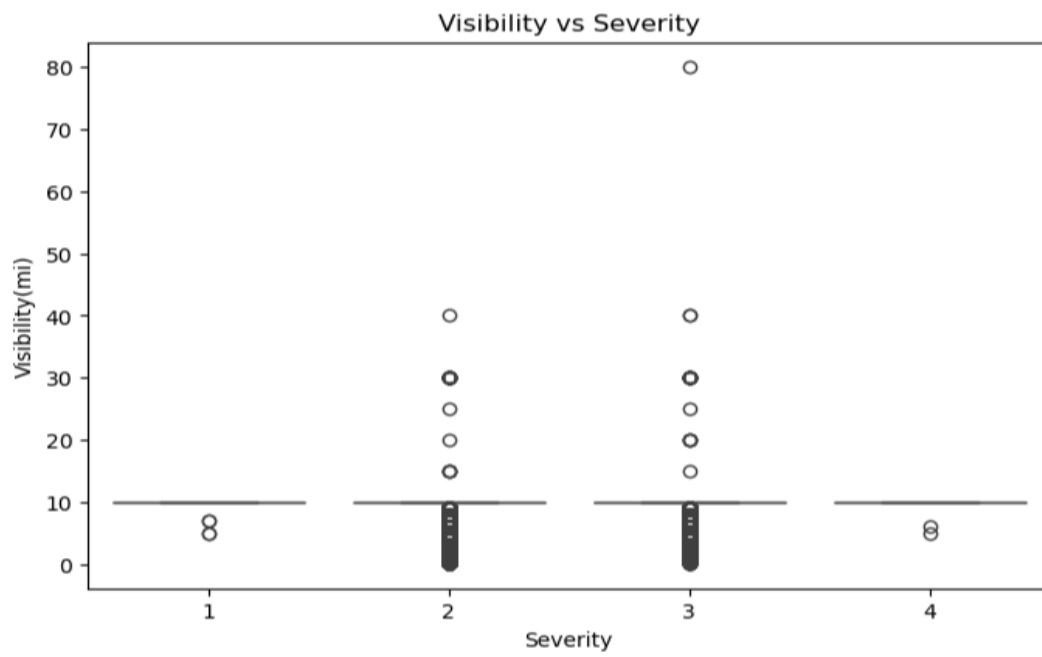
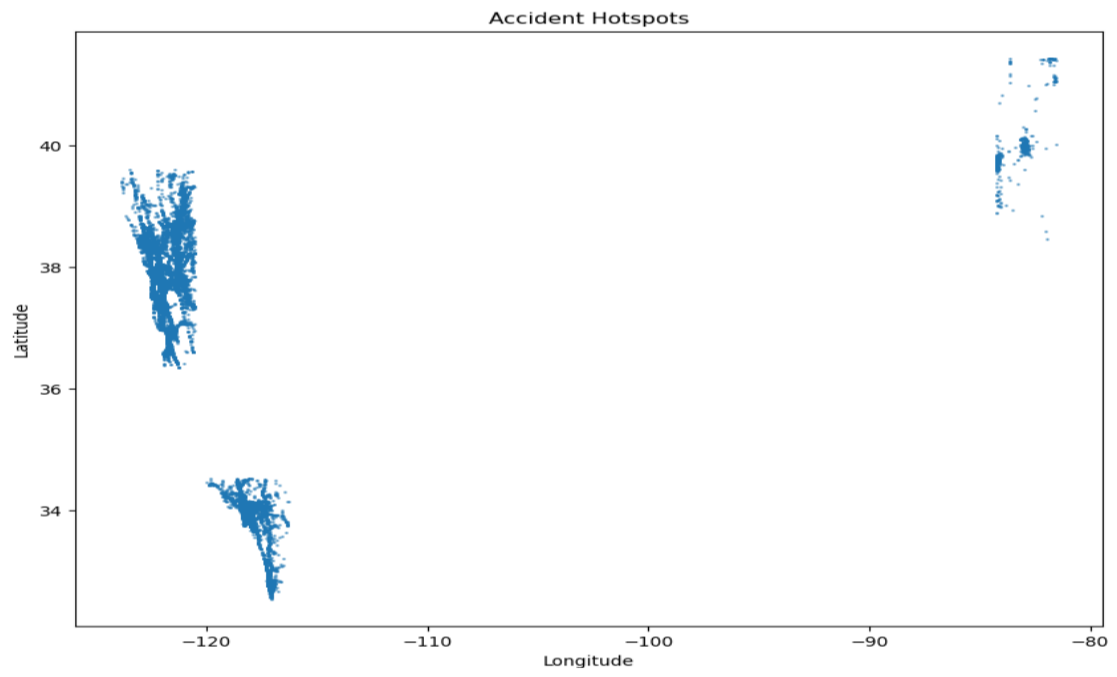


Figure 15



Key Findings

- Accident frequency varies across hours and days.
- Weather conditions influence accident occurrence.
- Certain states experience higher accident counts.
- Environmental factors affect accident severity.
- Visual analytics help identify accident hotspots.

Conclusion

Exploratory analysis of the US Accidents dataset reveals meaningful temporal, geographical and environmental patterns that can support road safety planning and traffic management.